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REVIEW



Using sequential pharmacotherapy for the treatment of osteoporosis: an update of the literature

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ABSTRACT

Introduction: Osteoporosis, which is characterized by compromised bone density and heightened susceptibility to fractures, is a substantial public health concern, especially among the aging population. Underdiagnosis, undertreatment, and therapy non-adherence contribute to its impact. Anabolic and dual-action agents like teriparatide, abaloparatide, and romosozumab have emerged as effective treatments, allowing rapid gains in bone mineral density (BMD) and reducing fracture risk. However, administering treatments in the correct order is paramount, with an 'anabolic first' approach gaining traction for patients at high risk of fractures. This strategy involves starting anabolic therapies, followed by antiresorptive agents as maintenance therapy. It is important to note that the effectiveness of anabolic agents differs between treatment-naïve and previously treated patients: tailored treatment approaches are therefore necessary. This comprehensive strategy adheres to clinical guidelines, emphasizing individualized care, early intervention, and patient-centered management to mitigate the burden of osteoporosis and enhance patients' quality of life.

Area covered: The aim of this review is to summarize recent evidence on the sequential treatment of osteoporosis and to provide recommendations on the best treatment strategies.

Expert opinion: Effective treatments, such as anabolic agents, are key in high-risk patients, who require an 'anabolic first' approach. Sequential therapy, specifically tailored to a patient's history, can help to optimize prevention and management of fractures.

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Anabolic agents; dual-action agent; high risk of fracture; osteoporosis; sequential treatment

1. Introduction

Osteoporosis is a silent chronic disease characterized by reduced bone density and abnormal architecture, predisposing patients to an increased risk of fractures [1]. Among Caucasian adults aged 50 years and over living in the U.S.A., 50% of women and 20% of men will experience an osteoporotic fracture in their lifetime [2]. These osteoporotic fractures represent a significant burden as they are associated with a loss of independence and increased morbidity and mortality [3,4], thus constituting a major public health concern. Unfortunately, osteoporosis is frequently underdiagnosed and therefore undertreated, even in patients at high risk of fractures [5–8]. Furthermore, treatment compliance is often suboptimal due to concerns about side effects, the route of administration of certain drugs, treatment costs and lack of patient education and awareness regarding the disease and its treatment [9–12].

Although several effective anti-osteoporotic treatments are currently available, choosing the best cost-effective individualized therapeutic approach is still challenging.

Due to potential issues with intolerance, limitation on duration of use and the possibility of treatment failure with certain agents [13,14], it is important to consider sequential treatment with different agents. Questions regarding the most appropriate sequence in which to administer different treatments has been raised during the past few years. Although anabolic and dual-action agents appear to be superior to antiresorptive agents in reducing

vertebral fracture risk, especially in patients at very high fracture risk [15–18], their action tends to wear off rapidly after discontinuation, therefore treatment needs to be maintained with antiresorptive agents in order to sustain BMD [19–21].

International guidelines recommend selecting the therapies according to the patients' level of risk [16,22–25]. The choice of the initial treatment will influence the patients' follow-up and may potentially impact the pharmacological approach that will subsequently be recommended. It is therefore fundamental to plan a patient's follow-up from the outset. Several national and international guidelines propose sequential therapy to achieve the best fracture risk reduction for osteoporotic patients [19,23,25].

The aim of this review is to summarize recent evidence on sequential treatment and to provide an expert recommendation about the best treatment strategy.

2. Body of review

Antiestoporosis agents are classified into three categories according to their main action on bone turnover: antiresorptive agents targeting osteoclasts (OCs), including bisphosphonates, selective estrogen receptor modulators (SERMs) and monoclonal antibodies against the receptor activator of nuclear factor kappa-B ligand (RANKL, denosumab), anabolic agents targeting osteoblasts (OBs), such as teriparatide and abaloparatide and a dual-

Article highlights

- Anabolic and dual action agents, such as teriparatide, abaloparatide and romosozumab, constitute the new frontier in osteoporosis treatment as they allow rapid gains in bone mineral density (BMD), dramatically reducing the risk of fracture.
- Current international guidelines support an 'anabolic first' approach for patients considered at very high risk of fracture.
- As anabolic drugs can only be prescribed for a limited period, maintenance treatment with antiresorptive agents is recommended.
- Although anabolic and dual-action agents are often prescribed as second line agents after antiresorptive treatment failure, their effectiveness is greater in naïve patients: this evidence will lead to changes in clinical practice in the near future.

action agent, romosozumab, which has the double effect of stimulating bone formation by targeting OBs and inhibiting bone resorption by targeting OCs through stimulation of the WNT- β catenin pathway.

2.1. Antiresorptive agents

2.1.1. SERMs

SERMs are synthetic pharmacological compounds that bind estrogen receptors. They act as agonists or antagonists of estrogen receptor depending on the target tissue [26]. SERMs that have been approved for the treatment of osteoporosis include raloxifene [27] and bazedoxifene [28].

These drugs decrease the formation and the lifespan of OCs and stimulate the activity of OBs within bones [29]. In addition to the direct effects on bone cells, estrogens directly influence calcium metabolism by increasing its absorption in the gut and decreasing its renal excretion [30].

2.1.2. Bisphosphonates

Bisphosphonates are the oldest and most widely used anti-resorptive agents. They are derivatives of inorganic pyrophosphate. Like their natural analogue, bisphosphonates have high affinity for bone as they bind to hydroxyapatite crystals and suppress bone resorption, promoting apoptosis of OCs [31]. They are characterized by a long duration of action, thanks to their affinity with hydroxyapatite: this allows them to maintain their antifracture effect even after discontinuation, as they remain active in bone and inhibit the activity of OCs when they are released from bone matrix during bone resorption [32]. According to different studies, the effectiveness of bisphosphonates in reducing vertebral fracture risk ranges from 41% to 72% [22,33–37].

With regards to non-vertebral fractures, zoledronate, risenedronate and alendronate are all effective at reducing the risk of fractures [34,36,37]. However, ibandronate has been shown to be effective only in reducing the risk of vertebral fractures [33].

2.1.3. Denosumab

Denosumab is a human monoclonal antibody which selectively binds to RANKL, preventing it from activating its receptor RANK, that is expressed by OCs and their precursors, consequently inhibiting the formation, function and survival of these cells [38]. Denosumab is effective at increasing BMD

and reducing fracture incidence compared to placebo [20]. Even when compared head-to-head with zoledronate [39], ibandronate [40], alendronate [41] and risenedronate [42] denosumab induces a greater suppression of bone turnover and greater increase of BMD at all skeletal sites, both in patients pre-treated with oral or intravenous bisphosphonates and in bisphosphonate-naïve patients [39,43]. However, there is currently no evidence suggesting the superiority of denosumab over bisphosphonates reducing the risk of fractures [44].

2.2. Anabolic agents

2.2.1. Teriparatide and abaloparatide

These synthetic peptides are respectively analogues of parathyroid hormone and of parathyroid hormone-related protein, which stimulate OBs recruitment and activity, increasing bone mass and improving bone quality [45–47]. These agents have been shown to be effective in treatment of osteoporosis. Daily injections of teriparatide increase BMD of the spine and reduce the risk of new vertebral and non-vertebral fractures by 65% and 69% respectively, when compared with placebo [48]. Abaloparatide significantly reduced the risk of new vertebral fractures by 86% versus placebo [49].

In several European countries, teriparatide is reimbursed as a second-line agent in patients with severe osteoporosis, particularly in the presence of multiples vertebral fractures. Patients undergoing treatment with teriparatide have therefore often already undergone treatment with an antiresorptive agent, and this may blunt their clinical response to teriparatide, as we will discuss later in the paper as discussed later in this paper.

Abaloparatide is commercialized in the US and has been recently approved by the European Medicines Agency (EMA) [50] for the treatment of osteoporosis.

2.3. Dual-action agents

2.3.1. Romosozumab

Romosozumab, the only agent in this group, is a monoclonal antibody inhibiting the action of sclerostin (SOST), a down-regulator of bone formation and an up-regulator of bone resorption [51]. SOST is a glycoprotein mainly produced by osteocytes in response to mechanical stimuli. In the absence of stimuli, for instance in case of prolonged immobilization or absence of gravity during space travel, SOST secretion is increased and bone loss occurs. On the other hand, weight bearing exercises reduce the expression of SOST and bone formation is enhanced. SOST is a competitor for the Wnt proteins, preventing them from activating an enzymatic cascade which finally results in increased bone formation and decreased bone resorption. By binding SOST, romosozumab avoids the interaction between sclerostin and its receptor, thus stimulating bone formation and inhibiting bone resorption [52].

In the FRAME study, romosozumab reduced fracture risk by 73% compared with placebo [53]. When compared with alendronate, romosozumab is more effective at improving BMD and bone strength, as calculated by the finite element analysis (FEA) method based on CT datasets [17].

Figure 1 summarizes the targets of the main anti-osteoporosis agents.

2.4. Who needs treatment?

In patients who have never experienced an osteoporotic fracture, the BMD-based threshold for the diagnosis of osteoporosis based on the T-score (T score ≤ -2.5 standard deviations) [54] has several limitations which reduce its utility in identifying patients requiring treatment. Although BMD measurement alone is highly specific for the diagnosis of osteoporosis, it has a low sensitivity [54]. The majority of fractures (60–70%) occur in patients with a T-score higher than -2.5 SD [55]. Therefore, additional fracture risk assessments should be used to guide decisions regarding treatment; one of the widely used tools for risk assessment is the FRAX tool, a computer-based algorithm which calculates a 10-year probability of developing major osteoporotic fractures (spine, pelvis, forearm or humerus) and hip fractures. The FRAX tool integrates eight clinical risk factors (prior fragility fracture, parental hip fracture, smoking, systemic glucocorticoid use, excess alcohol intake, body mass index, rheumatoid arthritis and other causes of secondary osteoporosis) which, in addition to age and gender, contribute to a 10-year fracture risk estimation, allowing a stratification of patients into low, intermediate and high risk categories [56].

The FRAX tool can also integrate BMD values, thus improving fracture risk prediction.

The use of the FRAX tool as fracture risk assessment tool is recommended by several international guidelines [16].

However, the threshold that determines the need for treatment varies between different countries. Some countries, such as the U.S.A. use fixed probability thresholds as their intervention thresholds regardless of gender and age. With this

approach a fixed threshold of 20% FRAX 10-year probability of a major osteoporotic fracture is used [57].

Other countries, mainly in Europe, use age-dependent thresholds [16].

Finally, in the UK, the National Osteoporosis Guideline Group guideline (NOGG) recommends a combination of age-dependent and fixed thresholds [58]. We therefore suggest that clinicians follow their national guidelines to make decisions regarding thresholds for starting treatment based on whether their patients are classified as low, intermediate or high risk. An algorithm suggesting treatment decision scheme according with Kanis et al [59] is shown in Figure 2.

However, an internationally accepted rule is that all patients who have already had an osteoporotic fracture will require treatment.

2.5. How to choose the appropriate treatment option?

In order to have a tailored approach, it is important to consider the pharmacological features of each class of drugs, including their effects on other organ systems, their side effect profiles and their individual effects on bone metabolism. Table 1 summarizes the agents currently used for osteoporosis, with their administration modality and their main indications.

To select the right treatment for the right patient, the side effect profile of each agent must be taken into account. For example, SERMs enhance the protective effects of estrogens on the uterus and on breast tissue [60] and they represent therefore an attractive treatment option for patients at high

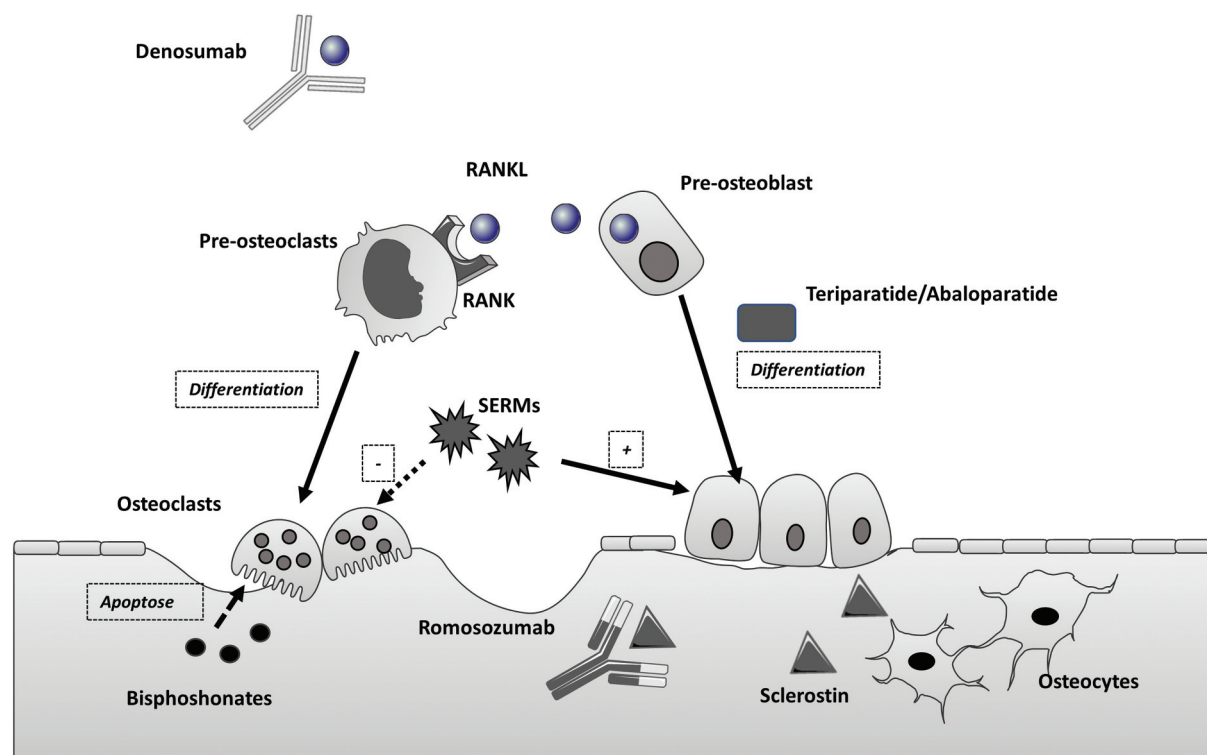


Figure 1. The scheme shows the mechanism of action of the different antiosteoporosis agents (SERMs = selective estrogen receptor modulators; RANKL = receptor activator of nuclear factor kappa-B ligand; RANK = receptor activator of nuclear factor kappa-B).

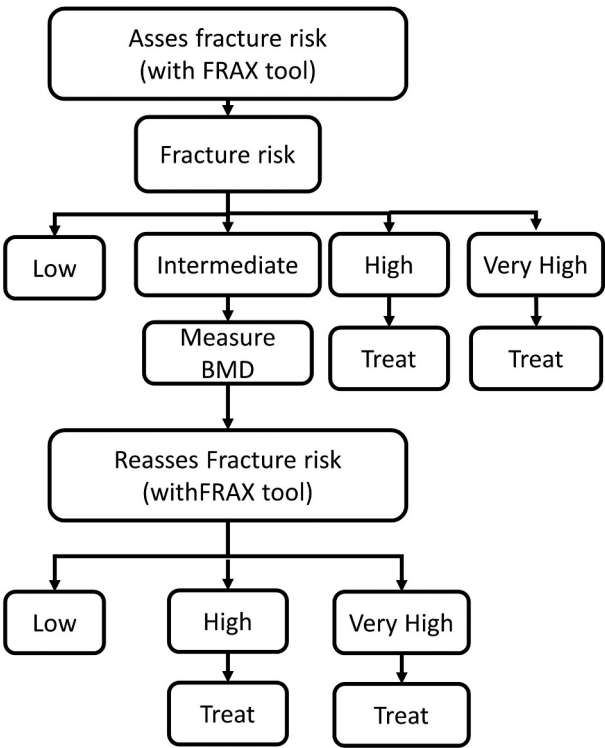


Figure 2. Algorithm to stratify patients according to their risk of fractures. Adapted from [59] Kanis et al., with permission.

risk of breast cancer [61]. However, they are associated with an increased risk of venous thromboembolism [62,63] and are therefore contraindicated in patients at high risk of thrombosis, such as those experiencing prolonged immobilization or those who have active neoplastic disease.

Regarding bisphosphonates, a large variety of formulations are available. Compliance may be improved by selecting once-weekly oral or intravenous regimens. Oral bisphosphonates have a low bioavailability, which requires them to be taken while fasting. The most common side effect experienced with bisphosphonates is upper gastrointestinal irritation. As a result, patients are required to remain upright for at least 30 minutes after their ingestion [64]. These gastrointestinal side effects affect 20–30% of users and are a common reason for drug discontinuation [65]. The most common side effect

related to intravenous bisphosphonates, which occurs in 1 out of 3 patients, is the development of a flu-like syndrome [36]. Bisphosphonates are contraindicated if the creatinine clearance is lower than 35 ml/min [66,67]. Furthermore, hypocalcemia must be corrected prior to introducing a bisphosphonate due to their propensity to cause hypocalcemia [68,69]. Very rare side-effects of long-term treatment with bisphosphonates include osteonecrosis of the jaw [70,71] and atypical femoral fractures [72]. It is therefore essential that patients are reevaluated every five years to assess whether the treatment should be continued or not and a drug holiday should be considered if the fracture risk is less than 20% over 10 years [73,74]

Denosumab is administered subcutaneously every 6 months [75]. It shares some of the side effects of bisphosphonates, including the risk of developing hypocalcemia, osteonecrosis of the jaw and atypical femoral fracture [20,76]. Similarly to bisphosphonates, osteonecrosis of the jaw and atypical fractures are very rare. Unlike bisphosphonates, denosumab does not need to be adapted in the presence of renal failure or in patients undergoing dialysis [77]. Moreover, unlike bisphosphonates, denosumab is not incorporated into bone and therefore its effects on BMD are reversed upon its discontinuation [44,78]. It is important to note that the sudden interruption of denosumab may cause a rebound effect which is associated with an increased risk of fractures, as discussed later in this paper.

Teriparatide and abaloparatide are administered by subcutaneous injection once a day. The use of teriparatide is limited to a maximum of two years by the FDA as a result of a study demonstrating an increased risk of developing osteosarcomas in rodents treated with high doses of teriparatide [79]. Reassuringly, a subsequent post-marketing study of patients treated with teriparatide has not found an increased risk for osteosarcoma risk compared to general population [80,81].

Romosozumab is also administered via subcutaneous injection once a month. The ARCH trial showed an increased risk of cardiovascular events in patients receiving this agent, thus romosozumab should not be initiated in patients with myocardial infarction or stroke in the preceding year [82] and benefits/risks of therapy should be consider in patients with other cardiovascular risk factors.

Table 1. Treatment options available.

Molecule	Dosage	Indication
Raloxifene	60 mg/day orally	Postmenopausal osteoporosis
Bazedoxifene	20 mg/day orally	Postmenopausal osteoporosis
Alendronate	10 mg/day or 70 mg/week orally	Postmenopausal osteoporosis and male osteoporosis, glucocorticoid induced osteoporosis
Risedronate	5 mg/day or 35 mg/week orally	Postmenopausal osteoporosis, male osteoporosis, Paget disease of bone, glucocorticoid induced osteoporosis
Zoledronate	5 mg/year intravenously	Postmenopausal osteoporosis, male osteoporosis, Paget disease of bone, glucocorticoid induced osteoporosis
Ibandronate	150 mg/month orally	Postmenopausal women
	3 mg/3 months intravenously	
Denosumab	60 mg/6 months subcutaneously	Postmenopausal women and male osteoporosis, aromatase inhibitor treatment, antiandrogenic treatment, glucocorticoid induced osteoporosis
Teriparatide	20 mcg/day subcutaneously, max 24 months	Postmenopausal women and male osteoporosis, glucocorticoid induced osteoporosis
Abaloparatide	80 mcg/day subcutaneously, max 18 months	postmenopausal osteoporosis
Romosozumab	210 mg/month subcutaneously, max 12 months	postmenopausal osteoporosis

Similarly to bisphosphonates and denosumab, romosozumab can cause hypocalcemia [83].

Romosozumab-induced hypocalcemia was reported as a rare side effect (incidence of <0.1%) in both the ARCH and FRAME trials and in 1% of patients in the STRUCTURE trial. To minimize this side effect, all these trials recommend optimizing vitamin D3 supplementation prior to romosozumab administration [15,82,84]. Clinicians should therefore be aware of the symptoms related to hypocalcemia, and should have a low threshold to identify and treat it in patients receiving romosozumab.

Table 2 summarizes the main advantages and side effects of the different anti osteoporosis drugs.

2.6. Does the order of treatment matter?

Given that osteoporosis is a chronic disease, most patients will likely be exposed to various agents. Patients previously treated with bisphosphates should be transitioned to anabolic agents if treatment failure with first-line drugs occurs or if there is an increased risk of fractures after treatment cessation. Furthermore, antiresorptive agents are required as maintenance therapy following treatment with denosumab, anabolic and dual-action agents because their effects tend to wear off rapidly after discontinuation [19–21].

The following paragraphs will deal with specific clinical situations.

2.6.1. What if antiresorptive agents aren't enough?

Because anabolic agents are only reimbursed as second-line treatments in most countries, most patients to whom these agents are prescribed will have been previously treated with an anti-resorptive agent, usually bisphosphonates.

Despite this clinical practice, there is evidence that shows that anabolic agents and dual-action agents are more effective in treatment-naïve patients, compared with previously treated patients.

The STRUCTURE trial showed that patients treated with romosozumab or teriparatide for a year had less significant BMD increases if they had had prior long-term treatment with bisphosphonates (generally over 6 years) when compared with treatment naïve patients [82].

With regards to romosozumab, study by Cosman et al. [85] showed that larger gains in BMD were achieved when this agent was used before, rather than after, antiresorptive agents such as alendronate and denosumab. Since BMD is a strong surrogate for bone strength and fracture risk, this analysis supports the notion that an initial treatment with romosozumab followed by an antiresorptive agent will result in greater efficacy than the reverse order.

With regards to teriparatide, Ettinger et al. [86] demonstrated that there is no statistical difference in the increase of bone formation markers when comparing treatment-naïve patients with those previously treated with raloxifene. However, previous treatment with alendronate and risedronate [87] blunted the effect of teriparatide on BMD increase. A hypothesis to explain these findings is that the suppression of bone turnover caused by antiresorptive agents reduces the availability of OBs and pre-OBs, which are the target of anabolic agents, diminishing the effect of these drugs.

Although the effect of teriparatide is blunted by a prior treatment with bisphosphonates, especially during the 6 months immediately following the transition from antiresorptive agent to anabolic agent, this combination still allows a statistically significant increase in BMD after 24 months cycle [88,89]. Therefore, treatment in this order, although not optimal, could be considered.

However, if the antiresorptive agent used before teriparatide is denosumab, patients will lose BMD [90]. This effect on bone metabolism in patients discontinuing denosumab may relate to the stimulation of a large pool of dormant osteoclast precursors by teriparatide, in patients in whom RANKL

Table 2. Advantages and side effects of the anti-osteoporosis drugs.

Molecule	Pros	Cons
SERMs	Breast cancer prevention	- Increased risk of thromboembolism - Increased risk of stroke - Hot flushes
Bisphosphonates	- Many galenic forms available - Reduction of hypercalcemia (in patients with bone metastasis)	- Low bioavailability for oral forms - Upper gastrointestinal irritation for oral form - Flu-like syndrome - Increased risk of atypical fracture - Increased risk of osteonecrosis of the jaw - Contraindicated in case of kidney failure. - Hypocalcemia
Denosumab	- No need to adapt the dose in case of renal failure - Good safety profile for 10 years	- Rebound effect at discontinuation with increase of vertebral fracture risk (pay attention to patient's compliance) - Increased risk of atypical fracture - Increased risk of osteonecrosis of the jaw - Increased risk of infection at the site of injection
Anabolics	Very rapid effect on BMD	- Contraindication in case of active neoplastic disease - Increased risk of infection at the site of injection - Pain in the extremities - Dizziness - Hypercalcemia
Romosozumab	Very rapid effect on BMD	- Increased risk of cardiovascular events - Increased risk of stroke - Increased risk of infection at the site of injection

inhibition has been sustained for two years. Therefore, switching from denosumab to an anabolic agent is not recommended as it may increase the risk of fractures.

2.6.2. *What if you need to stop denosumab*

The use of denosumab for 10 consecutive years is associated with an ongoing gain in BMD while maintaining an acceptable tolerance and safety profile [20]. Whether it is still beneficial to continue denosumab beyond 10 years is unclear, since studies have shown that the risk of rare side effects such as osteonecrosis of the jaw may increase if used for over 10 years, without definitive evidence of benefit on fracture risk. As denosumab discontinuation is associated with a 'rebound effect' characterized by increased bone turnover, decreased BMD and increased vertebral fracture risk [91,92], another osteoporosis treatment should be started if denosumab is discontinued [91,93]. In the DATA trial, women who received denosumab maintained the BMD that they had achieved at the end of the trial 15 months after the end of the study if they were taking another antiresorptive agent and no patient in this group experienced vertebral fractures [94].

The choice of selecting the right treatment sequence is of paramount importance: in the retrospective study by Burkhardt et al., the protective effect of bisphosphonates on vertebral and non-vertebral fractures was reduced if denosumab was given before, rather than after the bisphosphonates [95]. Similarly, the DAPS (Denosumab Adherence Preference Satisfaction) study came to the same conclusion: more gains in BMD were achieved when participants received denosumab after alendronate [96,97]. These experimental studies confirm the results of previous observational studies on the same topic. In order to minimize the increase in bone turnover which occurs when denosumab is discontinued, it is preferable to prescribe the more potent bisphosphonates such as alendronate or zoledronate [91,98]. Since bisphosphonates are deposited on areas of bone resorption and denosumab blocks bone resorption, these agents should only be administered as maintenance therapy when the rebound effect has already begun, usually months after the discontinuation of denosumab. This is particularly true for the infusion of zoledronic acid, which should be delayed to 6 months after denosumab discontinuation [91]. As previously discussed, switching from denosumab to teriparatide is not recommended as this transition leads to reduction in BMD [90]. Switching to romosozumab could be considered, knowing that the clinical response to this drug will be blunted in these patients [85].

2.6.3. *What is the best treatment after teriparatide and abaloparatide?*

Anabolic agents can be used for two years. Thereafter, the fracture risk must be reevaluated and if it remains high another treatment should be prescribed. The BMD gain achieved during treatment and the fracture risk reduction are maintained even after teriparatide is discontinued, provided patients are switched to an antiresorptive agent [99,100]. Some clinical trials have shown that denosumab is particularly effective at inducing an

additional gain in BMD compared to bisphosphonates following treatment with teriparatide [101]. Alendronate has also shown to be effective in decreasing fracture risk after discontinuation of abaloparatide [102,103].

There is currently no available data regarding the clinical efficacy and safety of transitioning from an anabolic agent to a dual-action agent. A recent study found that a high baseline serum procollagen 1 intact N-terminal propeptide (P1NP), a marker for bone formation, was significantly and independently associated with increased BMD in hips and femoral necks after 12 months of romosozumab treatment in patients with treatment-naïve postmenopausal osteoporosis. Conversely, low plasma levels of P1NP, indicative of reduced osteoblastic activity, may predict a poor response to treatment [104].

We can therefore speculate that the administration of an anabolic agent before romosozumab could maximize its effect in these patients. Further studies are needed to clarify this topic.

2.6.4. *... and after romosozumab?*

Romosozumab, provides rapid and substantial gains in bone mineral density at the spine and hip, but its administration is authorized for one year only. As this agent is often reserved for patients at very high risk of fracture, the question of subsequent treatment arises.

According to the FRAME study, the magnitude of initial BMD gain achieved in women receiving romosozumab was maintained, when this agent was followed by denosumab [105]. These results were subsequently confirmed by the randomized controlled trial by McClung et al. [106].

In the ARCH study, treatment with romosozumab for 12 months was followed by alendronate for 12 months. This allowed participants to maintain the BMD increase previously achieved and was associated with a reduction in the risk of vertebral fractures [53]. Similarly, a zoledronate follow-on regimen can maintain the BMD gain achieved with romosozumab treatment [107]. From these studies we can deduce that switching from romosozumab to an antiresorptive agent is an effective option to ensure the maintenance of BMD, thus lowering the risk of fracture.

In the study by Kendler et al. [108], a sequential therapy involving 24 months of romosozumab, followed by 12 months of denosumab and further 12 months of romosozumab resulted in even further gains in BMD at the hip and lumbar spine during the romosozumab re-challenge.

Consequently, a second course of romosozumab could be an option for patients remaining at very high risk after completing the first cycle of sequential treatment, however additional evidence is needed to support these findings.

Table 3 summarizes the clinical trials included in this review on sequential treatment.

In Figure 3 we propose a decision algorithm to help clinicians in selecting the initial treatment, the timing for risk of fracture reassessment and potential treatment sequences.

Table 3. Main clinical trials regarding sequential treatment discussed in this review.

Study	Design	Participants	Duration	Molecules	Population
DAPS [97]	Randomized, open-label, crossover study	250, mostly Caucasian	2 years (12 months Denosumab and 12 months Alendronate)	Denosumab 60 mg/6 months and Alendronate 70 mg/week	Naïve postmenopausal women (T score between – 2.0 and – 4.0)
EFOS [99]	Prospective, observational study	1649 in total, 907 who had a maintenance treatment after Teriparatide. Mostly Caucasian.	18 months of Teriparatide and 18 months of follow up	Teriparatide 20 mcg and other osteoporosis treatment	Postmenopausal women with osteoporosis (mean T score LS – 3.3, mean T score hip – 2.61)
ExFOS [100]	Prospective, observational study	1454, all Caucasian	Max 24 months of Teriparatide, 18 months of follow up	Teriparatide 20 mcg/day and other osteoporosis treatment	Postmenopausal women and men with severe osteoporosis, cortico-induced osteoporosis
ACTIVEext [102]	Randomized controlled trial (ABL vs PBO) followed by open-label trial with ALN	1139, mostly Caucasian, who completed ACTIVE	18 months of ABL follow by 24 months of ALN	Abaloparatide 80 mcg/day and Alendronate 70 mg/week	Postmenopausal women (mean T score LS – 2.91 in the PCB arm et – 2.88 in the ABL arm at the beginning of ACTIVE)
FRAME [105]	Randomized, double-blind, placebo-controlled, parallel-group trial	7180, mostly Hispanic (43%)	12 months romosozumab, 12 months Denosumab (extended to 24 months)	Romosozumab 210 mg/month and Denosumab 60 mg/6 months	Postmenopausal women (T-score LS or hip between – 2.5 and – 3.5)
ARCH [17]	Randomized, active-controlled, double-blind study	4093	12 months of Romosozumab and 24 months of ALN	Romosozumab 210 mg/month and Alendronate 70 mg/week	Postmenopausal women with severe osteoporosis (T score LS < – 2.5 or hip – 2.0 AND ≥ vertebral fracture or hip fracture in the previous 24 months)

3. Conclusion

Anabolic and dual action agents allow a fast and effective BMD gain, ensuring prompt protection against osteoporotic fractures. Given that treatments with anabolic and dual-action agents are limited to 12–24 months and that efficacy will vanish once treatment is stopped, the real potential of these treatments is that their effect on BMD and fracture risk can be preserved with a maintenance treatment once the anabolic/dual-action agent is interrupted. The American association of clinical endocrinologists' guideline proposes that treatment with an anabolic or dual-action agent should be followed by a bisphosphonate or by denosumab to prevent decline in bone density [25]. European and Asian guidelines are aligned in proposing an anabolic-first approach in patients at high risk of fractures [19,23]. These points should be considered when selecting the initial and subsequent management of patients with osteoporosis.

4. Expert opinion

Osteoporosis is a significant health concern characterized by compromised bone density and structure, resulting in heightened susceptibility to fractures. This insidious condition, often undetected until fractures occur, is a significant burden, especially for the aging population. Osteoporotic fractures trigger a cascade of consequences, ranging from loss of independence to increased morbidity and mortality. Therefore, osteoporosis must be considered as an important public health concern.

An important issue is the frequent underdiagnosis and undertreatment of osteoporosis. Even high-risk patients are not always identified and managed appropriately. This issue is a result of the subtle nature of the disease, patients'

misconceptions, and healthcare system inefficiencies. Furthermore, therapy adherence often wanes due to concerns about adverse effects, complex dosing regimens, financial constraints, and inadequate patient education.

Fortunately, a large spectrum of effective anti-osteoporotic treatments exists, although selecting the most optimal and cost-effective approach remains challenging. The emergence of anabolic and dual-action agents – exemplified by teriparatide, abaloparatide, and romosozumab – has introduced a new therapeutic dimension. These agents offer rapid gains in bone mineral density and marked reductions in fracture risks. Particularly noteworthy are anabolic agents' potential to stimulate osteoblast activity, fostering bone formation and resistance to fractures.

When making treatment decisions, the sequential nature of therapy requires careful attention. The order of treatments matters, especially considering the pivotal shift toward anabolic and dual-action agents. These compounds exhibit superior efficacy in reducing vertebral fracture risks, especially among high-risk patients. However, their potency vanishes after cessation, requiring ongoing therapy with antiresorptive agents to sustain BMD gains and fracture risk reduction.

Clinical guidelines endorse a selection of treatment aligned with the patient's fracture risk level. This not only influences the initial treatment but also subsequent steps, demanding a comprehensive outlook from clinicians right from the outset. Sequential therapy has gained traction as an avenue to maximize fracture risk reduction in osteoporotic patients. This involves initiating treatment with one class of drugs and transitioning strategically to another to capitalize on their respective benefits.

An 'anabolic first' approach has gained prominence for patients with an increased fracture risk. This involves commencing therapy with anabolic or dual-action agents to take

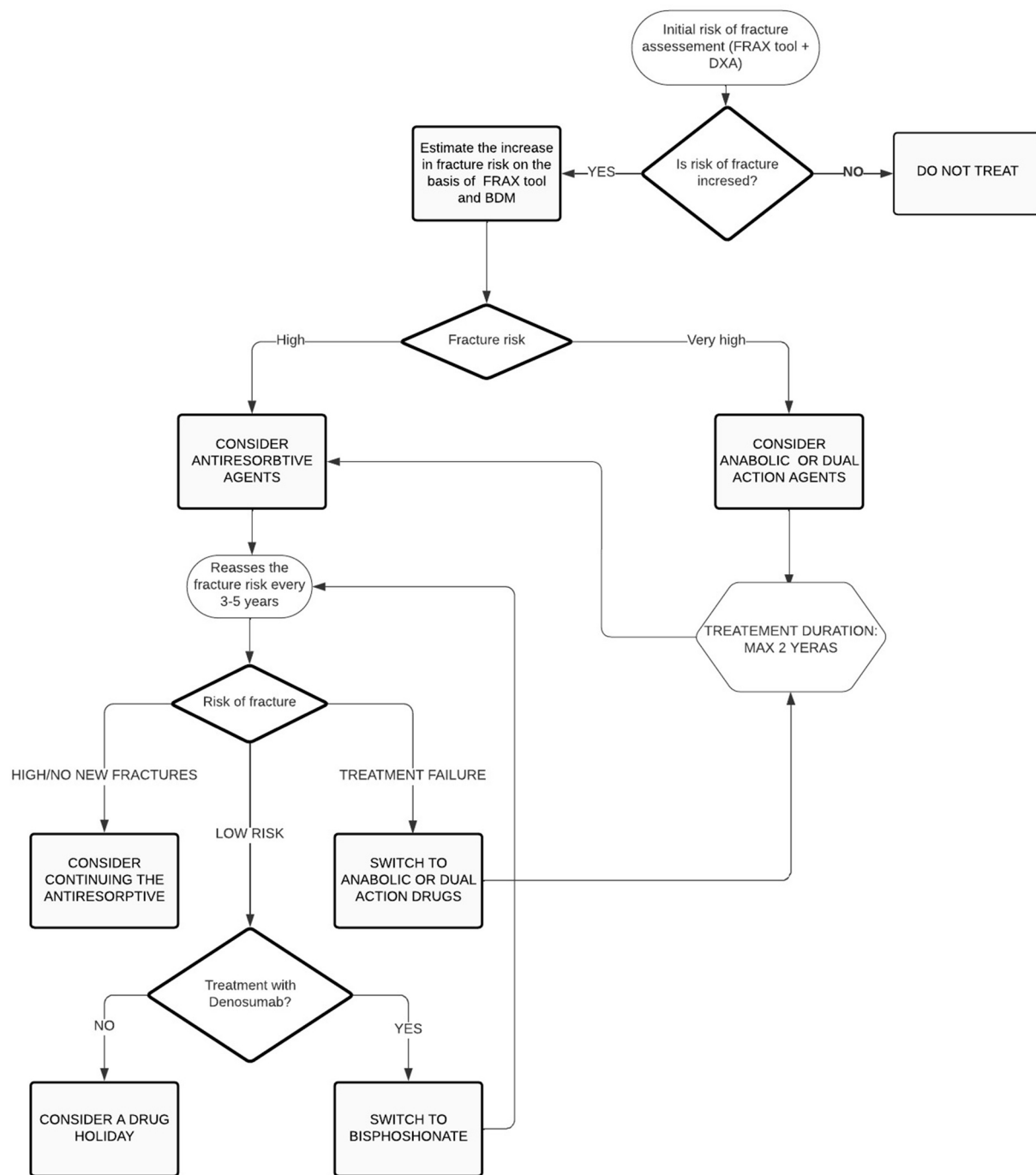


Figure 3. Decision algorithm concerning the choice of initial treatment, risk of fracture reassessment and potential treatment sequences.

advantage of their rapid BMD augmentation capabilities. These agents offer a substantial initial boost in bone density, rapidly reducing the fracture risk. However, the pivotal role of these agents extends beyond their standalone applications. The subsequent treatment course, typically transitioning to antiresorptive agents, underscores their lasting impact. By preventing bone resorption, antiresorptive agents complement the earlier gains, ensuring a more sustained and comprehensive fracture risk reduction strategy.

A nuanced consideration is the variability in treatment response between treatment-naïve patients and those who have previously undergone treatment with antiresorptive

agents. Anabolic and dual-action agents exhibit superior efficacy in patients without prior treatment, requiring a reconsideration of treatment sequencing. This insight suggests that the clinical community must recalibrate treatment strategies to accommodate patient history.

In conclusion, the evolving landscape of osteoporosis treatment emphasizes sequential therapy and the strategic integration of anabolic and dual-action agents. The 'anabolic first' approach holds promise, offering rapid BMD gains and fracture risk reduction. However, the journey does not end with the initial gains. The subsequent selection of antiresorptive agents determines the longevity and sustainability of the

therapeutic benefits. This iterative process, guided by fracture risk assessment, patient history, and treatment characteristics, embodies a comprehensive approach to managing osteoporosis. By aligning treatment sequencing with patient needs, clinicians can usher in a new era of fracture prevention and improved quality of life for those grappling with osteoporosis.

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